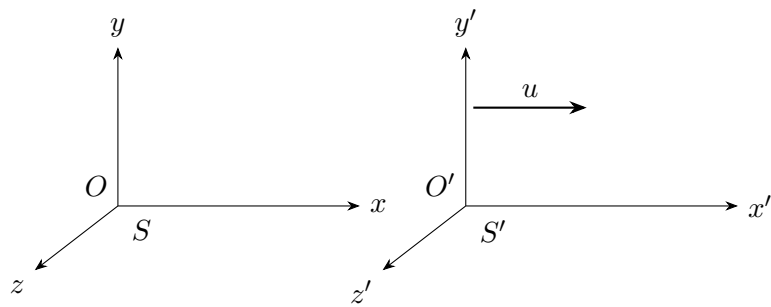


Special Relativity Notes

Time Dilation, Length Contraction and the Lorentz Transformation

04/09/26

Thought Experiment (Gedanken experiment)



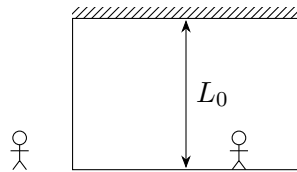
Using the Galilean transformation:

$$c' = c - u$$

Therefore length and time are not absolute. They will change with the speed of the observer.

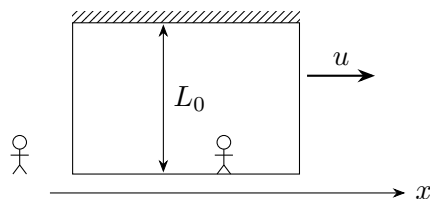
1 Time in a Relativistic Frame

Case 1 — car at rest

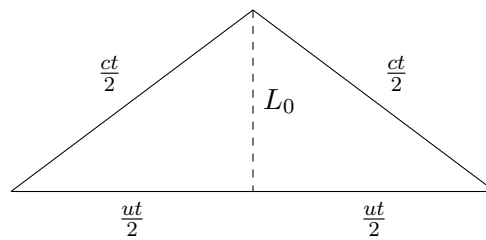


$$t_0 = \frac{2L_0}{c} \quad (\text{when the car is at rest})$$

Case 2 — car moving with speed u



Q. What will the observer sitting outside the car see?



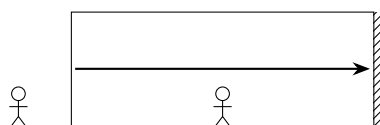
$$\begin{aligned} \left(\frac{ct}{2}\right)^2 &= L_0^2 + \left(\frac{ut}{2}\right)^2 \\ \Rightarrow t^2(c^2 - u^2) &= 4L_0^2 \\ \Rightarrow t^2 &= \frac{4L_0^2}{c^2 \left(1 - \frac{u^2}{c^2}\right)} \end{aligned}$$

$$t = \frac{t_0}{\sqrt{1 - \frac{u^2}{c^2}}}$$

- If $u = 0$: $t = t_0$
- If $u < c$: $t \geq t_0$

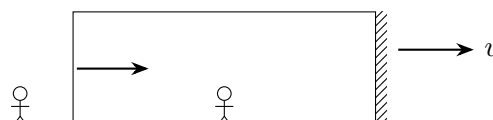
2 How Length Changes with the Speed of the Observer

Case 1 — car at rest



$$t_0 = \frac{2L_0}{c} \quad (\text{when the car is at rest})$$

Now the car moves with speed v :



Q. What will the observer see once the car has started moving relativistically?

Forward journey, the time taken:

$$t_1 = \frac{L + vt_1}{c}$$

Backward journey, the time taken:

$$t_2 = \frac{L - vt_2}{c}$$

Total time:

$$\begin{aligned} t = t_1 + t_2 &= \frac{L + vt_1}{c} + \frac{L - vt_2}{c} \\ &= \frac{L}{c} + \frac{vt_1}{c} + \frac{L}{c} - \frac{vt_2}{c} \end{aligned}$$

Exercise — show that $L = L_0 \sqrt{1 - v^2/c^2}$

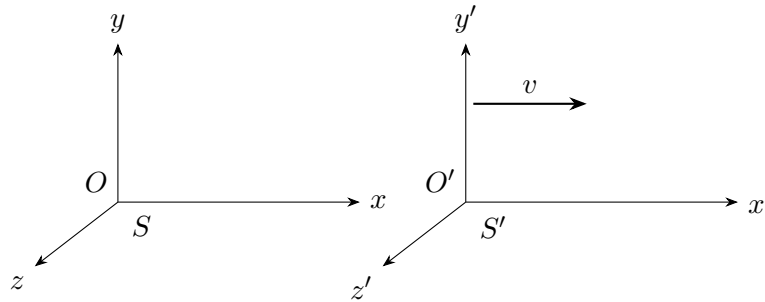
From $t_1 = \frac{L}{c - v}$ and $t_2 = \frac{L}{c + v}$, and using the time-dilation result $t = \frac{2L_0}{c\sqrt{1 - \frac{v^2}{c^2}}}$:

$$\begin{aligned} \frac{2L_0}{c\sqrt{1 - \frac{v^2}{c^2}}} &= \frac{2L}{c} + \frac{vL}{(c - v)c} - \frac{vL}{(c + v)c} \\ \text{(multiplying by } c) \quad \frac{2L_0}{\sqrt{1 - \frac{v^2}{c^2}}} &= 2L + vL \frac{(c + v) - (c - v)}{c^2 - v^2} \\ &= 2L + 2L \left(\frac{1}{\frac{c^2}{v^2} - 1} \right) = 2L \left(\frac{\frac{c^2}{v^2}}{\frac{c^2}{v^2} - 1} \right) = \frac{2L}{1 - \frac{v^2}{c^2}} \end{aligned}$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

- If $v = 0$: $L = L_0$
- If $v < c$ (e.g. $v = 0.9c$): $L = L_0 \sqrt{0.19} \Rightarrow L < L_0$

3 Lorentz Transformation



Lorentz transformation

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad y' = y, \quad z' = z, \quad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

In the rest frame, the coordinates of the event are (x, y, z, t) .

In the moving frame, the coordinates of the event are (x', y', z', t') .

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & -v\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{v\gamma}{c^2} & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}$$

$$\gamma \rightarrow \text{boost factor}, \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$